COL 703/7203

19th Oct

$\forall x\,(\phi \wedge \psi) \vdash \forall x\phi \wedge \psi$

assuming that x is not free in ψ.

$\forall x\,(\phi \wedge \psi)$ premise

$\exists x\, Q(x)$

x_0 arbitrary

$(\phi \wedge \psi)[x_0/x]$

$\phi[x_0/x] \wedge \psi[x_0/x]$ (Same as above)

$\phi[x_0/x]$ $\wedge e$

$\forall x\,\phi$

$\forall x\,(\phi \wedge \psi)$

	$(\phi \wedge \psi)[x_0/x]$	$\forall e$
	$\phi[x_0/x] \wedge \psi[x_0/x]$	same as above
	$\psi[x_0/x]$	$\wedge e$
m	ψ	because x is not free in ψ
	$\forall x\,\phi \wedge \psi$	$\wedge i$ $\quad l, m$

$\forall x (\phi \to \psi) \vdash \exists x \phi \to \psi$

assuming x is not free in ψ.

1. $\exists x \phi$ assumption
2. $\forall x (\phi \to \psi)$ premise
3. x_0 $\phi[x_0/x]$ assumption
⋮
$(\phi \to \psi)[x_0/x]$ $\forall e$
$\phi[x_0/x] \to \psi[x_0/x]$ same as above
$\psi[x_0/x]$ MP
ψ
ψ $\exists e$ 1, 3...

(assuming x is not free in ψ)

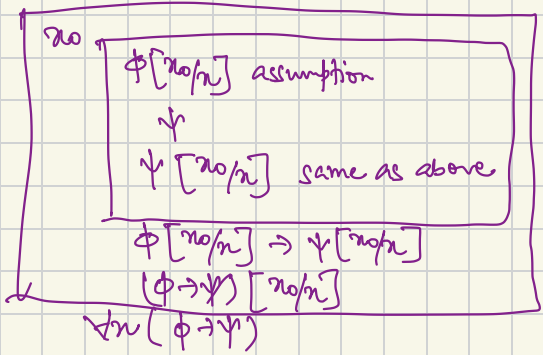
$\exists x\,\phi \to \psi \;\vdash\; \forall x\,(\phi \to \psi)$

$\exists x\,\phi \to \psi$ premise

$\exists x\,\phi \;\vee\; \neg\exists x\,\phi$ (LEM)

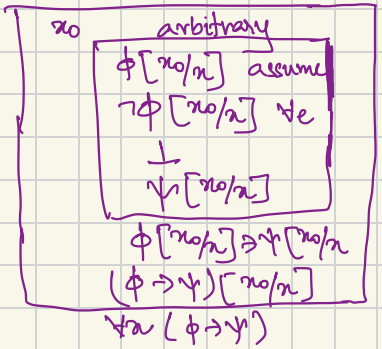
$\exists x\,\phi$

ψ MP



$\neg\exists x\,\phi$
$\vdots$
$\forall x\,\neg\phi$

this was done in the last class



$$\exists x \exists y \phi \vdash \exists y \exists x \phi$$

$$\phi[x_0/x][y_0/y]$$

Same as

$$\phi[y_0/y][x_0/x]$$

$\forall x \exists y \phi \quad \dashv \quad \exists y \forall x \phi$

Exercise

Exercise

$\forall x \exists y \left(P(x) \vee Q(y) \right) \dashv$

$\exists y \forall x \left(P(x) \vee Q(y) \right)$

COL703/7203: Logic for Computer Science (Jul-Nov 2026)

Predicate Logic – Semantics, Normal Forms

Kumar Madhukar

madhukar@cse.iitd.ac.in

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Semantics

$\forall x \forall y \ (P(x, y) \rightarrow \exists z \ (\neg(z = x) \wedge \neg(z = y) \wedge P(x, z) \wedge P(z, y)))$

- suppose variables take values from real numbers
- P(x, y) represents $x < y$

Semantics

$\forall x \forall y \ (P(x, y) \rightarrow \exists z \ (\neg(z = x) \wedge \neg(z = y) \wedge P(x, z) \wedge P(z, y)))$

- suppose variables take values from real numbers
- P(x, y) represents $x \leq y$

Semantics

$$\forall x \forall y \ (P(x, y) \rightarrow \exists z \ (\neg(z = x) \wedge \neg(z = y) \wedge P(x, z) \wedge P(z, y)))$$

- suppose variables take values from natural numbers
- P(x, y) represents $x < y$

σ-structure[1]

Given a signature σ, a *σ-structure* (or *assignment*) $\mathcal{A}$ consists of:

- a non-empty set $U_{\mathcal{A}}$ called the *universe* of the structure;
- for each k-ary predicate symbol P in σ, a k-ary relation $P_{\mathcal{A}} \subseteq \underbrace{U_{\mathcal{A}} \times \cdots \times U_{\mathcal{A}}}_{k}$;
- for each k-ary function symbol symbol f in σ, a k-ary function, $f_{\mathcal{A}} : \underbrace{U_{\mathcal{A}} \times \cdots \times U_{\mathcal{A}}}_{k} \to U_{\mathcal{A}}$;
- for each constant symbol c, an element $c_{\mathcal{A}}$ of $U_{\mathcal{A}}$;
- for each variable x an element $x_{\mathcal{A}}$ of $U_{\mathcal{A}}$.
- for each atomic proposition (nullary predicate) a T/F assignment

[1] Based on James Worrell's notes

Evaluating terms

- For a constant symbol c we define $\mathcal{A}[\![c]\!] \stackrel{\text{def}}{=} c_{\mathcal{A}}$.
- For a variable x we define $\mathcal{A}[\![x]\!] \stackrel{\text{def}}{=} x_{\mathcal{A}}$.
- For a term $f(t_1, \ldots, t_k)$, where f is a k-ary function symbol and $t_1, \ldots, t_k$ are terms, we define $\mathcal{A}[\![f(t_1, \ldots, t_k)]\!] \stackrel{\text{def}}{=} f_{\mathcal{A}}(\mathcal{A}[\![t_1]\!], \ldots, \mathcal{A}[\![t_k]\!])$.

Satisfaction relation

1. $\mathcal{A} \models P(t_1, \ldots, t_k)$ if and only if $(\mathcal{A}[\![t_1]\!], \ldots, \mathcal{A}[\![t_k]\!]) \in P_{\mathcal{A}}$.
2. $\mathcal{A} \models (F \wedge G)$ if and only if $\mathcal{A} \models F$ and $\mathcal{A} \models G$.
3. $\mathcal{A} \models (F \vee G)$ if and only if $\mathcal{A} \models F$ or $\mathcal{A} \models G$.
4. $\mathcal{A} \models \neg F$ if and only if $\mathcal{A} \not\models F$.
5. $\mathcal{A} \models \exists x\, F$ if and only if there exists $a \in U_{\mathcal{A}}$ such that $\mathcal{A}_{[x \mapsto a]} \models F$.
6. $\mathcal{A} \models \forall x\, F$ if and only if $\mathcal{A}_{[x \mapsto a]} \models F$ for all $a \in U_{\mathcal{A}}$.

If we are working in first-order logic with equality then we additionally have

7. $\mathcal{A} \models t_1 = t_2$ if and only if $\mathcal{A}[\![t_1]\!] = \mathcal{A}[\![t_2]\!]$.

Exercise

Consider a signature σ with a constant symbol 0, a unary function symbol s, and a unary predicate symbol P.

Is $P(0) \wedge \forall x\ (P(x) \rightarrow P(s(x))) \wedge \exists x\ \neg P(x)$ satisfiable?